

Report — Task 2: Analysis Report

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Architectural Changes

```
class GreenNet(nn.Module):
    """
    Two Conv-ReLU-MaxPool blocks, followed by Average Pooling and a single FC layer.
    """
    def __init__(self, in_channels=3, num_classes=4, **kwargs):
        super().__init__()
        self.conv1 = nn.Conv2d(in_channels, 16, kernel_size=3, padding=1)
        self.bn1 = nn.BatchNorm2d(16)
        self.conv2 = nn.Conv2d(16, 32, kernel_size=3, padding=1)
        self.bn2 = nn.BatchNorm2d(32)
        self.relu = nn.ReLU()
        self.pool = nn.MaxPool2d(2)
        self.gap = nn.AdaptiveAvgPool2d(1)
        self.fc = nn.Linear(32, num_classes)

    def forward(self, x):
        x = self.pool(self.relu(self.conv1(x)))
        x = self.pool(self.relu(self.conv2(x)))
        x = self.gap(x)
        x = x.flatten(1)
        return self.fc(x)
```

The key design choices, compared to the three baseline models:

- **Only two convolutional stages** (16 then 32 channels), against AlexNet's five conv layers (up to 256 channels), VGG16's 13 conv layers (up to 512 channels), and ResNet18's 8 residual blocks (up to 512 channels).
- **Global Average Pooling (GAP) instead of a flatten-into-dense-layer classifier head.** This is the single biggest parameter saving in the design. AlexNet and VGG16 both flatten their final feature maps into a multi-thousand-dimension vector before their first `Linear` layer (AlexNet: 3072 → 1024, VGG16: 2048 → 1024) — those two layers alone account for millions of parameters. GreenNet's GAP collapses the final feature map straight to a 32-length vector, so its entire classifier is one `Linear(32, num_classes)` layer.
- **BatchNorm after every conv**, which helps the network train stably despite having far less depth to smooth out noisy gradients.

Efficiency Verification Matrix

Averaged across all four datasets (cells, chest, lesions, orgs):

Model	Avg. Train Time (s)	Avg. Peak Train Memory (MB)	Avg. Inference Latency (ms/sample)	Avg. Accuracy
GreenNet	16.66	71.7	0.880	70.95%
AlexNet	46.40	186.3	2.317	84.86%
VGG16	330.60	854.4	6.070	83.06%
ResNet18	649.41	1349.3	8.346	83.46%

GreenNet trains roughly 3x faster than AlexNet, 20x faster than VGG16, and 39x faster than ResNet18, while using **2.6x less memory than AlexNet, 12x less than VGG16, and 19x less than ResNet18**. Inference latency follows the same pattern: GreenNet responds in under a millisecond per sample, compared to over 8ms for ResNet18 — a roughly 9x difference.

Accuracy Trade-Off

Dataset	GreenNet Accuracy	Best Baseline Accuracy	Gap
cells	69.07%	95.57% (VGG16)	-26.50
chest	75.96%	89.58% (ResNet18)	-13.62
lesions	67.13%	73.34% (AlexNet)	-6.21
orgs	71.63%	91.77% (ResNet18)	-20.14